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Title
Understanding Strapdown Integration

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Answer
The **Strapdown Integration (SDI)** algorithm in all Xsens' products is extremely important for the performance of orientation estimates.

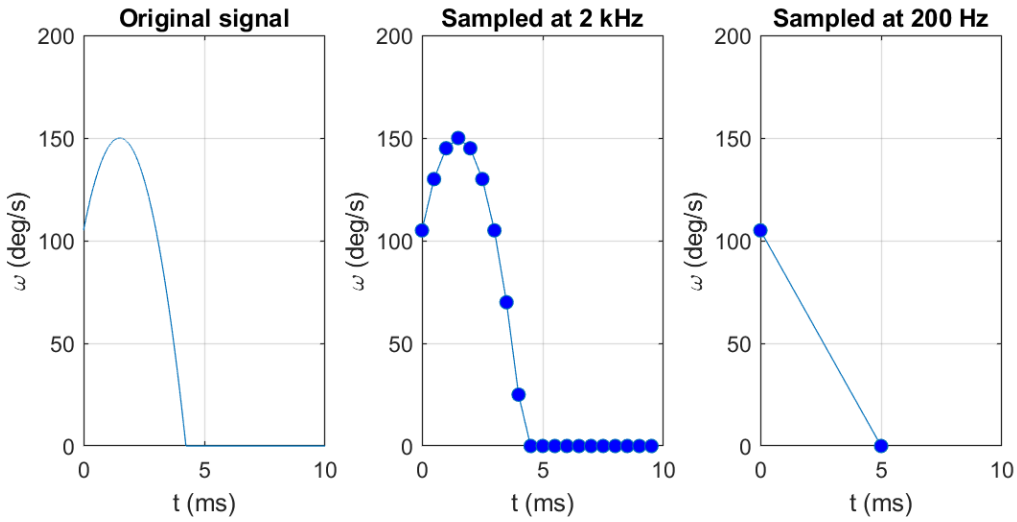
Strapdown integration is the process of sampling and integrating sensor data in order to determine the change in location and/or orientation of a device or vehicle. The term strapdown indicates that the measurement device is strapped onto the application, in contrast with the use of gimbals for determining orientation. The use of sensors strapped onto an application in combination with a microprocessor eliminates issues such as gimbal lock, and allows determining position and orientation at a much higher accuracy and data rate. The resulting data, in combination with a starting point or reference, can provide a full navigation input for a large variety of unmanned systems. Example applications include robotic vehicles and drones.

The Xsens MTi product series consist of a 3D sensor assembly combined with a microprocessor. The on-board SDI algorithm performs highly accurate computation of the integrated quantities of angular velocity (Δq) and acceleration (Δv) with coning and sculling error compensation. The Xsens MTi sensor modules make use of 3D MEMS accelerometer and gyroscope assemblies for this purpose. The generated SDI quantities (orientation and velocity increments) form the main input for the attitude estimation filters of the VRU, AHRS and GNSS/INS products in the MTi product range as well as custom navigation solutions developed by customers.

The most crucial part of the signal processing chain is when the analog sensor signals are transformed (sampled) to the digital domain. For the MTi 10/100 series, this is done at 2000 Hz. The signal is then passed through calibration and strapdown integration including downsampling. The MTi products can be configured to output the integrated Δq and Δv quantities as well as the corresponding Rate of Turn and Accelerometer data at various output frequencies, from as low as 1 Hz all the way up to 400 Hz for the MTi 10/100-series. Because of Xsens' advanced SDI algorithm, all filtering is taken care of by the on-board processor, in a dynamical way. This means that all the information of higher frequencies is integrated into the output signal, even when the user selects a low output frequency.

Example 1: Capturing high bandwidth motion at a low output data rate

Consider a gyroscope signal ω that needs to be sampled and integrated in order to determine how far a vehicle has rotated (Δq). A small part of this movement is shown in the first plot below.



When sampling at a rate of 2 kHz (middle plot), nearly all information of the original signal is preserved because of the high sampling rate.

Finally, consider the same motion, but the gyroscope signal is now sampled and integrated at 200 Hz instead (right plot). Information will be lost, since the sample rate is too low to capture all dynamics.

Using trapezoidal integration[1], we can estimate the integrated quantity Δq over the full time window of 10 ms for each signal:

Original signal: Δq = 0.47636 deg

Sampled at 2 kHz: Δq = 0.47625 deg

Sampled at 200 Hz: Δq = 0.26250 deg

When using the Xsens SDI algorithm, all of the 2000 Hz signal will be integrated and then sampled down to the desired output frequency, i.e. 100 Hz. The Δq values resulting from that will contain the full motion involved, even though the output data rate would normally be too low to capture all dynamics.

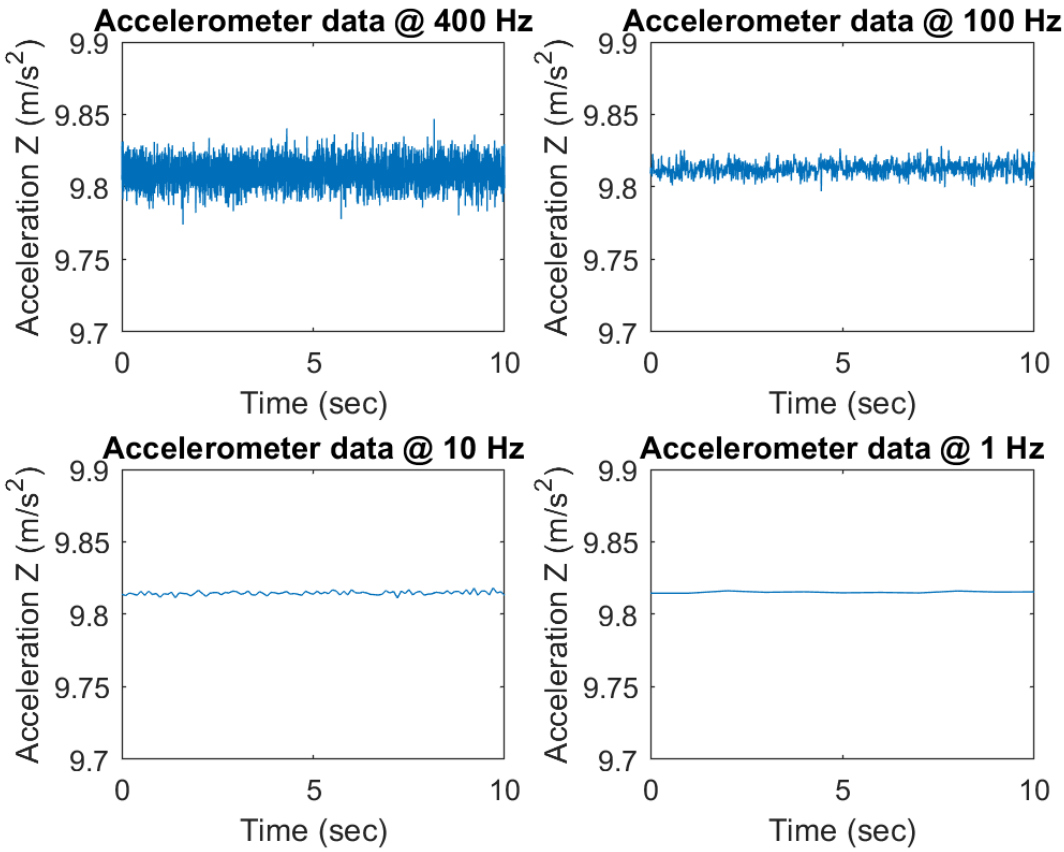
Xsens' SDI algorithm offers more benefits when using the Acceleration and Rate of Turn data outputs, computed from the strapdown integrated quantities, compared to simply sampling sensor data.

The noise density of the accelerometer and gyroscope data is irrespective of the output frequency: the noise density specifications are constant and can be found in the datasheets of the MTi products. However, if it is desired to have a lower RMS noise, it is possible to lower the output rate of the MTi. The SDI algorithm will then simply integrate the data over a longer period of time. This means that the RMS noise decreases.

Example 2a: Dynamic filtering of noise in Accelerometer/Gyroscope data

An MTi-300 was held motionless for 10 seconds and configured to output acceleration data at 400 Hz, 100 Hz, 10 Hz and 1 Hz. The difference in standard deviation is clearly visible in the plots below. The standard deviations that correspond to these four output data rates are:

- 400 Hz: 0.00903 m/s^2
- 100 Hz: 0.00485 m/s^2
- 10 Hz: 0.00132 m/s^2
- 1 Hz: 0.00056 m/s^2

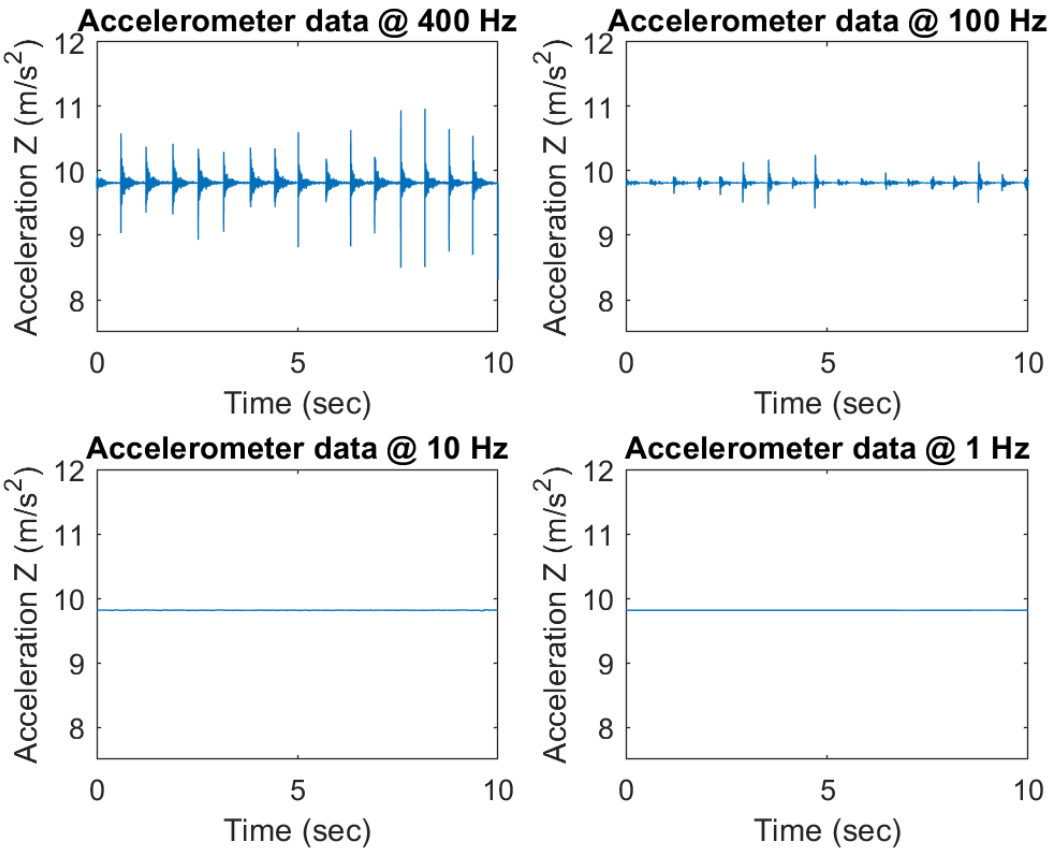


Besides the decrease of the RMS noise, lowering the output rate of the MTi will also mean that higher frequency components will not be visible in the output. This is useful in case you are only interested in lower bandwidths.

Example 2b: Dynamic filtering of vibrations in Accelerometer/Gyroscope data

An MTi-300 was held motionless on a table and configured to output acceleration data at 400 Hz, 100 Hz, 10 Hz and 1 Hz. The casing of the MTi was then tapped by hand, several times. The resulting z-axis acceleration plots are shown below. In the 400 Hz signal, the resulting damped vibrations can clearly be seen. The 100 Hz signal already shows a significant reduction of their intensity. In the plots for 10 Hz and 1 Hz, the vibrations were completely filtered out by the SDI algorithm. This is in contrast to when sensor data is sampled directly at 10 Hz or 1 Hz; the effect of the vibrations would still be visible in the output signal.

Note: *these four experiments were carried out individually, but at the same tapping frequency and intensity.*



For more information, technical details and performance tests of the MTi products, refer to the MTi Whitepaper: https://www.xsens.com/whitepaper-download/ (https://www.xsens.com/whitepaper-download/)

[1] https://nl.mathworks.com/help/matlab/ref/trapz.html (https://nl.mathworks.com/help/matlab/ref/trapz.html), not the actual integration method used in Xsens' SDI algorithm.

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